

Linear constraints

Basic solutions

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Optimization: principles and algorithms



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Basic solutions

The concept of vertex is important, but is a geometric concept

We follow up on the algebraic representation introduced in the last video

We introduce the concept of basic solutions.

Definition

Consider

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x and the indices form a **basic solution** if

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If

$$x_B = B^{-1}b \geq 0,$$

it is a **feasible basic solution**.

Equivalence

Theorem 3.40

$x^* \in \mathcal{P}$ is a vertex of $\mathcal{P}$ if and only if it is a feasible basic solution.

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Idea of the proof: sufficient condition

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- ▶ Consider k such that $x_k^* > 0$.
- ▶ Build a feasible direction such that $d_k = 1$ and $d_j = 0$ for j non basic.
- ▶ It is possible to find y and z along that direction such that $\exists 0 < \lambda < 1$ and

$$x^* = \lambda y + (1 - \lambda)z.$$

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- ▶ Calculate the basic and non basic parts of x , y and z .

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- ▶ Assume by contradiction that x^* is not a vertex.
- ▶ Consider y and z different from x^* such that $x^* = \lambda y + (1 - \lambda)z$.
- ▶ Calculate the basic and non basic parts of x , y and z .
- ▶ Show that $x = y = z$.

Summary

Basic solution: algebraic concept.

Equivalence with vertices.

It is important to perform the practice quizzes to understand concretely how it works.